

24CS31 — SEE Prediction Sheet

Laplace Transforms & Vector Space · III Semester B.E. CSE · Batch 2024

Built from: 4 distinct previous question papers (Apr 2023, Jun 2023 Make-up, Aug/Sep 2023 Backlog, Mar 2024 SEE) · every question transcribed and verified against the page image.

Filtered against: the official 24CS31 syllabus (Ramaiah Institute of Technology, curriculum AY 2025-26).

Back-test: the Tier 1 + Tier 2 list below scores **100 / 100** on all four papers when the recommended section is chosen. See §9.

Read this first — two honest caveats

- Two of your five uploads are the same paper.** SEE_25.pdf and IS131_MATH_SEE_2024_1.pdf are both the **SEMESTER END EXAMINATIONS - MARCH 2024** paper (one is a clean digital copy, the other a scan). They are identical question-for-question. So the evidence base is **4 papers, not 5** — and nothing here double-counts them.
- Almost nothing in these papers is out of your syllabus.** You asked me to rule out off-syllabus material; I checked all 4×10 questions against the 24CS31 unit list and only **one** item (Markov steady-state, Mar 2024 Q9b) is not literally named in the syllabus text — and it still appeared under your exact course title, so it stays in. The real gap runs the other way: **PCA and conic sections are in your syllabus and have never been asked.** That is the actual risk, and it is handled in §8.

§1 Your syllabus — the filter everything was judged against

Unit	Title	Content (official)	CO
I	Laplace Transforms	Definition; transforms of elementary functions; properties; existence conditions; transform of derivatives; transform of integrals; multiplication by t^n ; division by t ; evaluation of integrals by Laplace transforms; transform of a periodic function.	CO1
II	Application of Laplace Transforms	Unit-step function; unit-impulse function; inverse transforms; convolution theorem; solution of linear differential equations and simultaneous linear differential equations; engineering applications.	CO2
III	Vector Space & Linear Transformation	Vector space; linear combination and span; linearly independent / dependent vectors; basis and dimension; linear transformations; matrix of transformations; rotation about the origin; dilation, contraction and reflection; composition of matrix transformations; kernel and range; change of basis.	CO3
IV	Orthogonal Projections	Null space of A ; solving $Ax=0$ and $Rx=0$; complete solution to $Ax=b$; dimensions of the four subspaces; orthogonality of the four subspaces; projections; orthonormal bases and Gram-Schmidt; QR-factorization; least-squares approximations.	CO4
V	Applications of Eigenvalue Decomposition	Eigenvalues and eigenvectors; similarity and diagonalization; symmetric matrices; complex matrices; Hermitian and unitary matrices; positive definite matrices; SVD; PCA ; applications to linear recurrence relations; quadratic forms and conic sections .	CO5

Syllabus verdict on the 40 questions analysed

Verdict	Count	Detail
In syllabus	39 / 40	Every Unit I-IV question and all but one Unit V question maps directly onto a named syllabus line.
Borderline — keep	1 / 40	Markov steady-state (Mar 2024 Q9b, 5 m). Not named in the syllabus text, but it is an eigenvalue-decomposition application and it appeared under your exact course title. Cost to learn: ~15 minutes. Keep it.
Out of syllabus — drop	0 / 40	Nothing. The 2023 papers ran under the older title “Linear Algebra and Laplace Transforms”, but their content sits inside your five units without remainder.
In syllabus, never asked	—	PCA; conic sections; existence conditions of the Laplace transform; a direct linear-recurrence (Fibonacci-style) question. These are your blind spots — see §8.

§2 The SEE paper pattern — anatomy

Constant across all four papers: Max marks **100** · Duration **3 hours** · 10 questions · **“Answer one full question from each unit.”** There is no separate Part-A/Part-B, no compulsory question, and no internal choice inside a question

— the choice is at question level only.

Unit	Your choice	Structure — Mar 2024 (current, CS/IS/AI/AD/CY/CI31)	Structure — 2023 papers (older title)	Unit total
I	Q1 or Q2	a (02) + b (04) + c (07) + d (07)	a (06) + b (07) + c (07)	20
II	Q3 or Q4	a (02) + b (04) + c (07) + d (07)	a (06) + b (07) + c (07)	20
III	Q5 or Q6	a (06) + b (07) + c (07)	a (06) + b (07) + c (07)	20
IV	Q7 or Q8	Q7: a (06)+b (07)+c (07) Q8: a (10)+b (10)	a (10) + b (10)	20
V	Q9 or Q10	a (05) + b (05) + c (10)	a (10) + b (10)	20

The blueprint to plan against

Use the March 2024 shape. It is the only paper printed under your exact course name (*Laplace Transforms and Vector Space*) and your exact course code family (*CS/IS/AI/AD/CY/CI31*). The 2023 papers ran the older course. What changed in 2024:

- Units I and II gained a **2-mark definition** and a **4-mark short problem** — 6 nearly-free marks per unit, 12 across the paper.
- Unit V split from 10+10 into **5 + 5 + 10** — two short property-checks plus one big decomposition.
- Unit IV became **asymmetric**: Q7 is 6+7+7 (three smaller problems), Q8 is 10+10 (two heavy ones). Expect this to persist — it lets the examiner put QR and least-squares together in Q8.

Mark-weight per unit under the current blueprint

Unit	2 m	4 m	5 m	6 m	7 m	10 m	Total
I — Laplace Transforms	1	1	-	-	2	-	20
II — Applications of LT	1	1	-	-	2	-	20
III — Vector Space	-	-	-	1	2	-	20
IV — Orthogonal Projections	-	-	-	1*	2*	2*	20
V — Eigenvalue Decomposition	-	-	2	-	-	1	20

* Unit IV: Q7 gives 6+7+7, Q8 gives 10+10. You pick one shape.

§3 Frequency analysis — every topic, every paper

A = Apr 2023 · **M** = Jun 2023 Make-up · **S** = Aug/Sep 2023 Backlog · **N** = Mar 2024 SEE (newest, current syllabus). Cell shows the question slot it occupied.

Unit I — Laplace Transforms

Topic	A · Apr 23	M · Jun 23	S · Sep 23	N · Mar 24	Hits	Verdict
Proof: $L\{t^n f(t)\} = (-1)^n d^n F/ds^n$	Q2b (7)	Q1b (7)	Q2c (7)	Q1c (7)	4/4	LOCK
$L\{\sin \sqrt{t}\} / L\{\cos \sqrt{t} / \sqrt{t}\}$	Q2a (6)	Q1c (7)	Q2a (6)	Q1b (4)	4/4	LOCK
Laplace transform of a periodic function	Q1c (7)	Q2c (7)	Q1b (7)	Q1a (2) Q2d (7)	4/4	LOCK
Division by t: evaluate $L\{f(t)/t\}$	Q1b (7)	Q1a, Q2b	Q1a (6)	Q2c (7)	4/4	LOCK
Multiplication by t^n / first shifting — $L\{t e^{at} \text{ trig}\}$	Q2c (7)	Q2a (6)	Q1a, Q2b	Q1d, Q2c	4/4	LOCK
Evaluate $\int_0^\infty \dots dt$ using Laplace transforms	Q2c (7)	-	Q1c (7)	Q1d (7)	3/4	Very high
Transform of an integral $L\{\int_0^t \dots\}$	Q1a (6)	Q2a (6)	-	-	2/4	Medium
$L\{(\sqrt{t} \pm 1/\sqrt{t})^3\}$	Q1a (6)	-	-	Q2b (4)	2/4	Medium
Proof: $L\{f(t)/t\} = \int_s^\infty F(s)ds$	-	Q2b (7)	-	-	1/4	Low
Proof: $L\{(\sinh at)f(t)\}$	-	-	Q2b (7)	-	1/4	Low
2-mark definition (LT / periodic function)	-	Q1a(i)	-	Q1a, Q2a	2/4	New format

Unit II — Application of Laplace Transforms

Topic	A · Apr 23	M · Jun 23	S · Sep 23	N · Mar 24	Hits	Verdict
Heaviside / unit-step piecewise expression + LT	Q4b (7)	Q4b (7)	Q3b (7)	Q3c (7)	4/4	LOCK
Convolution theorem (apply, verify, or state & prove)	Q3b (7)	Q3b (7)	Q3a, Q4c	Q4c (7)	4/4	LOCK
Simultaneous ODEs by LT (incl. particle-on-curve)	Q3c (7)	Q4c (7)	Q3c (7)	Q4d (7)	4/4	LOCK
Solve a single ODE by Laplace transforms	Q4c (7)	Q3c (7)	Q4b (7)	-	3/4	Very high
Inverse LT of log(...)	Q4a (6)	-	Q4a (6)	Q3b (4)	3/4	Very high
Inverse LT — partial fractions / shifting / $\tan^{-1}$	Q3a (6)	Q3a, Q4a	-	Q4b (4)	3/4	Very high
2-mark definition (unit step / Dirac delta)	-	-	-	Q3a, Q4a	1/4	New format
Engineering application (LR circuit current)	-	-	-	Q3d (7)	1/4	New — watch

Unit III — Vector Space and Linear Transformation

Topic	A · Apr 23	M · Jun 23	S · Sep 23	N · Mar 24	Hits	Verdict
Composition reflection $\rightarrow$ rotation $\rightarrow$ dilation/contraction, + image of a point	Q5a (6)	Q6c (7)	Q6a (6)	Q6a (6)	4/4	LOCK
Transition matrix / change of basis	Q5c (7)	Q5c (7)	Q6c (7)	Q6b (7)	4/4	LOCK
Kernel & range + verify rank-nullity	Q6c (7)	Q6b (7)	Q5c (7)	Q6c (7)	4/4	LOCK
Prove T is linear + find images	Q5b (7)	Q5b (7)	Q5b (7)	Q5b (7)	4/4	LOCK

Linear combination / span / LI-LD / basis of a space	Q6a (6)	Q5a, Q6a	Q5a (6)	Q5a (6)	4/4	LOCK
Proof: $T(x) = Ax$ rotates x through θ	Q6b (7)	-	-	Q5c (7)	2/4	Medium

Unit IV — Orthogonal Projections

Topic	A · Apr 23	M · Jun 23	S · Sep 23	N · Mar 24	Hits	Verdict
QR factorization	Q8b (10)	Q8a (10)	Q7a (10)	Q8a (10)	4/4	LOCK
Least-squares solution of $Ax = b$	Q7b (10)	Q8b (10)	Q8b (10)	Q8b (10)	4/4	LOCK
Four fundamental subspaces / basis of $\text{col}(A)$ & $\text{nul}(A)$	Q7a (10)	Q7a (10)	Q7b (10)	Q7c (7)	4/4	LOCK
Complete solution to $Ax = b$	Q8a (10)	Q7b (10)	-	Q7a (6)	3/4	Very high
Gram-Schmidt orthonormal basis	-	-	Q8a (10)	-	1/4	Live risk
Orthogonal projection of y onto u	-	-	-	Q7b (7)	1/4	New — watch

Unit V — Applications of Eigenvalue Decomposition

Topic	A · Apr 23	M · Jun 23	S · Sep 23	N · Mar 24	Hits	Verdict
Orthogonally diagonalize a symmetric matrix	Q10a (10)	Q9a (10)	Q10a (10)	Q10c (10)	4/4	LOCK
Singular Value Decomposition (SVD)	Q10b (10)	Q10b (10)	Q9b (10)	Q9c (10)	4/4	LOCK
Positive definite check	Q9b(ii)	Q9b(ii)	Q10b(ii)	Q10a (5)	4/4	LOCK
Hermitian matrix check	Q9b(i)	Q9b(i)	Q10b(i)	-	3/4	Very high
Diagonalize and hence find A^n	Q9a (10)	Q10a(i)	Q9a (10)	-	3/4	Very high
Unitary matrix check	-	Q10a(ii)	-	Q10b (5)	2/4	Medium
Quadratic form → no cross-product terms / definiteness	-	-	Q10b(ii)	Q9a (5)	2/4	Medium
Markov steady-state vector	-	-	-	Q9b (5)	1/4	New — watch
PCA — <i>in syllabus, never asked</i>	-	-	-	-	0/4	BLIND SPOT
Conic sections — <i>in syllabus, never asked</i>	-	-	-	-	0/4	BLIND SPOT

§4 Section-pairing laws — which questions travel together

This is the part that decides **which of the two questions you attempt**. Each unit offers two questions and you answer one whole question — so what matters is not just *which topics appear*, but *which topics land in the same question*. These laws were extracted by cross-tabulating all 40 questions.

LAW 1 — Unit I 4/4

The **t^n -multiplication proof** and the **$\sin \sqrt{t} / \cos \sqrt{t}$** question are always in the same question.

Evidence — Apr 23: Q2a + Q2b • Jun 23: Q1b + Q1c • Sep 23: Q2a + Q2c • Mar 24: Q1b + Q1c

So what: That is **11 marks (7+4)** sitting in one question. Find that question in the hall — it is your Unit I answer.

LAW 2 — Unit I 4/4

The full **periodic-function problem** is always in the opposite question to the pair in Law 1.

Evidence — Apr 23: periodic Q1c ↔ pair Q2 • Jun 23: Q2c ↔ Q1 • Sep 23: Q1b ↔ Q2 • Mar 24: Q2d ↔ Q1

So what: You cannot get both. Prepare both anyway — the periodic one is 7 marks of pure formula.

LAW 3 — Unit II 4/4

The **Heaviside piecewise question** and an **ODE-type problem** (single ODE, simultaneous ODEs, or the circuit application) are always in the same question.

Evidence — Apr 23: Q4b + Q4c • Jun 23: Q4b + Q4c • Sep 23: Q3b + Q3c • Mar 24: Q3c + Q3d

So what: **14 guaranteed marks (7+7)** in one question. This is the single most valuable pairing on the paper.

LAW 4 — Unit II 3/4

The **convolution** question sits in the opposite question to the Heaviside one.

Evidence — Apr 23 ✓ • Jun 23 ✓ • Sep 23 ✗ (Sep 23 put convolution in both) • Mar 24 ✓

So what: So you will almost certainly have to choose between Heaviside+ODE and convolution. Take Heaviside+ODE (Law 3).

LAW 5 — Unit III 3/4

The **vector-space pair** — {linear combination / span / LI-LD / basis} and {prove T is linear + find images} — sit together in **Q5**.

Evidence — Jun 23: Q5a + Q5b • Sep 23: Q5a + Q5b • Mar 24: Q5a + Q5b • Apr 23 ✗

So what: Q5 is the “definitions and proofs” question. Easier if you are shaky on computation.

LAW 6 — Unit III 3/4

The **composition-of-transformations** question and the **transition-matrix** question sit together — and in the two newest papers the **kernel&range** question joins them.

Evidence — Apr 23: Q5a + Q5c • Sep 23: Q6a + Q6c • Mar 24: **Q6a + Q6b + Q6c** (all three) • Jun 23 ✗

So what: Mar 24's Q6 is three LOCK topics in one 20-mark question. **Target the question containing the composition problem.**

LAW 7 — Unit IV 4/4

The **four-subspaces / column-space & null-space** question is always in Q7 — the first question of Unit IV.

Evidence — Apr 23: Q7a • Jun 23: Q7a • Sep 23: Q7b • Mar 24: Q7c

So what: Perfectly consistent across four papers. If you want the four-subspaces marks, you must answer **Q7**.

LAW 8 — Unit IV 2/4 — but both newest

QR factorization + least-squares sit together in **Q8**.

Evidence — Jun 23: Q8a + Q8b • Mar 24: Q8a + Q8b • Apr 23 ✗ • Sep 23 ✗

So what: **20 marks from two LOCK topics in one question**. The most recent paper does exactly this. This is your Unit IV target — but see the decision card in §7.

LAW 9 — Unit V 3/4

Orthogonal diagonalization and **SVD** are in opposite questions.

Evidence — Jun 23 ✓ • Sep 23 ✓ • Mar 24 ✓ • Apr 23 ✗ (both in Q10)

So what: You get one 10-mark decomposition, not both. Prepare both — they share the $A^T A$ machinery.

LAW 10 — Unit V 4/4

The **matrix-property checks** (Hermitian / positive-definite / unitary) always travel together as a block — either as (i)/(ii) of one 10-marker or as the 5+5 pair.

Evidence — Apr 23: Q9b(i)+(ii) • Jun 23: Q9b(i)+(ii) • Sep 23: Q10b(i)+(ii) • Mar 24: Q10a + Q10b

So what: 10 marks of 10-minute work. Two definition-checks, no heavy algebra. Never skip these.

LAW 11 — Unit V 3/4

Orthogonal diagonalization travels with the property-check block.

Evidence — Jun 23: Q9a + Q9b • Sep 23: Q10a + Q10b • Mar 24: Q10a + Q10b + Q10c • Apr 23 ✕

So what: Combined with Law 10 this gives the Unit V target: **the question containing orthogonal diagonalization is a full 20.**

LAW 12 — Unit V 3/4

SVD travels with either “diagonalize and find A^n ” or the short quadratic-form / Markov questions.

Evidence — Jun 23: Q10a + Q10b • Sep 23: Q9a + Q9b • Mar 24: Q9a + Q9b + Q9c • Apr 23 ✕

So what: Your fallback question if the orthogonal-diagonalization matrix looks unfamiliar.

§5 TIER 1 — repeating with the SAME VALUES

What qualifies: the question has already been set **two or more times with numerically identical data**, and in most cases the most recent appearance is the March 2024 paper. These are not “topics” — they are specific problems with specific matrices and specific functions. Work each one end-to-end until you can reproduce it from memory. **Marks covered by this tier alone: 63 of 100.**

T1-01 Orthogonally diagonalize the symmetric matrix

Orthogonally diagonalize $A = \begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix}$

Unit: V — symmetric matrices | **Marks:** 10 | **Section:** Q9 or Q10 — the **c** part (10 m). Was **Q10c** in Mar 24, **Q9a** in Jun 23, **Q10a** in Apr 23. | **Appears alongside:** the positive-definite check and the Hermitian/unitary check (Laws 10 & 11) — that whole question is a clean 20 | **History:** 3 of 4 papers, identical matrix every time | **Note:** eigenvalues are **2, 3 and 6** — all distinct, so the eigenvectors come out mutually orthogonal on their own. Just normalise them to build P , then $P^TAP = D$

T1-02 Least-squares solution of $Ax = b$

Find the least-squares solution of $AX = b$ for $A = \begin{bmatrix} 1 & 3 & 5 \\ 1 & 1 & 0 \\ 1 & 1 & 2 \\ 1 & 3 & 3 \end{bmatrix}$ and $b = \begin{bmatrix} 3 \\ 5 \\ 7 \\ -3 \end{bmatrix}$

Unit: IV — least-squares approximations | **Marks:** 10 | **Section:** **Q8b** — it was Q8b in Mar 24 and Q8b in Jun 23. | **Appears alongside:** the QR factorization (Law 8) — together they are the full 20 of Q8 | **History:** 2 of 4 papers with identical A and b ; the same A was also the QR matrix in Apr 23 | **Note:** this matrix is the examiner’s favourite — it has appeared three times in four papers in some role

T1-03 Four fundamental subspaces — dimension and basis

Find the dimension and basis for the four fundamental subspaces of $A = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{bmatrix}$

Phrased in Jun 23 as “find a basis for $\text{col}(A)$ and $\text{nul}(A)$ ” — same matrix, same work.

Unit: IV — dimensions of the four subspaces | **Marks:** 7 (was 10 in the 2023 papers) | **Section:** **Q7** — always. Q7a in Apr 23, Q7a in Jun 23, Q7b in Sep 23, Q7c in Mar 24 (Law 7). | **Appears alongside:** the complete solution of $Ax = b$ and the orthogonal projection (that is the whole of Mar 24 Q7) | **History:** 3 of 4 papers, identical matrix | **Note:** row 3 = row 1, so rank = 2; dim of nul = 2, dim of left-nul = 1

T1-04 Proof — multiplication by t^n

If $L\{f(t)\} = F(s)$, prove that $L\{t^n f(t)\} = (-1)^n \frac{d^n}{ds^n} \{F(s)\}$, where n is a positive integer.

Unit: I — multiplication by t^n | **Marks:** 7 | **Section:** Q1 or Q2, the **c** part in the new format. Q1c in Mar 24, Q1b in Jun 23, Q2b in Apr 23, Q2c in Sep 23. | **Appears alongside:** the $\sin \sqrt{t} / \cos \sqrt{t}$ question — **always** (Law 1) | **History:** 4 of 4 papers, word-for-word identical | **Note:** the single most reliable question on the entire paper — proof by induction on n

T1-05 Positive-definiteness check

Check whether the following matrix is positive definite or not: $A = \begin{bmatrix} 1 & -2 & 1 \\ -2 & 4 & -2 \\ 1 & -2 & 1 \end{bmatrix}$

Unit: V — positive definite matrices | **Marks:** 5 (was part of a 10 in 2023) | **Section:** **Q10a** in Mar 24; was Q9b(ii) in Apr 23. | **Appears alongside:** the Hermitian or unitary check — the property-check block never splits (Law 10) | **History:** 2 of 4 papers with this exact matrix; the *topic* is 4 of 4 | **Note:** leading minors are 1, 0, 0 — it is **positive semi-definite, not positive definite**. Say so explicitly

T1-06 Singular Value Decomposition

Find the singular value decomposition (SVD) of $A = \begin{bmatrix} 1 & 1 \\ 3 & -3 \end{bmatrix}$

Unit: V — SVD | **Marks:** 10 | **Section:** the **c** part of Q9 or Q10. Q9c in Mar 24, Q10b in Jun 23. | **Appears alongside:** the quadratic form and Markov questions (Mar 24 Q9) or the diagonalize-and-find- A^n question (Jun 23 Q10) | **History:** 2 of 4 papers with this exact matrix; the *topic* is 4 of 4 | **Note:** $A^T A = \begin{bmatrix} 10 & -8 \\ -8 & 10 \end{bmatrix}$ with eigenvalues 18 and 2, so the singular values are $3\sqrt{2}$ and $\sqrt{2}$

T1-07 Transition matrix / change of basis

Consider the bases $B = \{(1, 2), (3, -1)\}$ and $B' = \{(3, 1), (5, 2)\}$ of R^2 . Find the transition matrix from B to B' . If u is a vector such that $[u]_B = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, find $[u]_{B'}$.

Unit: III — change of basis | **Marks:** 7 | **Section:** Q6b in Mar 24; Q6c in Sep 23. (Apr 23 and Jun 23 used Q5c with the standard basis as B' .) | **Appears alongside:** the composition-of-transformations question and the kernel&range question — the three-LOCK Q6 block (Law 6) | **History:** 2 of 4 papers with these exact bases and this exact u ; the *topic* is 4 of 4 | **Note:** back-to-back in the two most recent papers — strongest same-values signal in Unit III

T1-08 Heaviside representation + Laplace transform

Express the function $f(t) = \begin{cases} t^2, & 0 < t < 2, \\ 4t, & 2 < t < 4, \\ 8, & t > 4. \end{cases}$ in terms of the Heaviside function and hence find its Laplace transform.

Unit: II — unit-step function | **Marks:** 7 | **Section:** Q3c in Mar 24; Q4b in Apr 23. | **Appears alongside:** an ODE-type problem — single ODE, simultaneous ODEs, or the circuit application (Law 3). **14 marks in one question.** | **History:** 2 of 4 papers with this exact piecewise function; the *topic* is 4 of 4 | **Note:** the other two papers used $\cos t/\cos 2t/\cos 3t$ and $1/t/t^2$ — same method

T1-09 Inverse Laplace transform of a logarithm

Evaluate $L^{-1} \left\{ \log \frac{s^2 + 1}{s(s + 1)} \right\}$

Unit: II — inverse transforms | **Marks:** 4 (was 6 in the 2023 papers) | **Section:** Q3b in Mar 24; Q4a in Apr 23. | **Appears alongside:** the Heaviside question (Mar 24 Q3) — so it sits in the *same* question as your 14-mark Law 3 pair | **History:** 2 of 4 papers with this exact expression; a log-inverse appeared in 3 of 4 | **Note:** differentiate $F(s)$, then use $L^{-1}\{F'(s)\} = -t f(t)$

T1-10 Proof — the rotation operator

Show that a linear operator $T: R^2 \rightarrow R^2$ defined by $T(x) = Ax$ rotates a vector x through an angle θ about the origin if $A = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$

Unit: III — rotation about the origin | **Marks:** 7 | **Section:** Q5c in Mar 24; Q6b in Apr 23. | **Appears alongside:** the “prove T is linear” question and the vector-space basis question — the Q5 proof block (Law 5) | **History:** 2 of 4 papers, identical statement | **Note:** your escape hatch if the Q6 computational block looks ugly — it is a bookwork proof

T1-11 Convolution theorem — the standard inverse

Find $L^{-1} \left\{ \frac{s^2}{(s^2 + a^2)(s^2 + b^2)} \right\}$ using the convolution theorem.

Unit: II — convolution theorem | **Marks:** 7 | **Section:** Q3b in both Apr 23 and Jun 23. Mar 24 instead asked you to *verify* the theorem for $f_1 = t, f_2 = \cos t$ at Q4c. | **Appears alongside:** the opposite question to Heaviside (Law 4) — usually with a plain inverse-LT part and an ODE | **History:** 2 of 4 identical; the *topic* is 4 of 4 | **Note:** Sep 23 set the same thing with $a=1, b=2$. Learn the general form and both specialisations

T1-12 Hermitian matrix check

Determine whether the matrix $A = \begin{bmatrix} 3 & 7-4i & -2+5i \\ 7+4i & -2 & 3+i \\ -2-5i & 3-i & 4 \end{bmatrix}$ is Hermitian.

Unit: V — complex / Hermitian matrices | **Marks:** 5 (was half of a 10 in 2023) | **Section:** the property-check block — Q9b(i) in Apr 23, Q10b(i) in Sep 23. | **Appears alongside:** the positive-definite check — they are the (i) and (ii) of the same part (Law 10) | **History:** 2 of 4 papers with this exact matrix; the *topic* is 3 of 4 | **Note:** it **is** Hermitian — the diagonal is real and every a_{ij} equals the conjugate of a_{ji} . Write out A^* in full and compare entry by entry

T1-13 Given $L\{\sin \sqrt{t}\}$, show the $\cos \sqrt{t}$ result

Given $L\{\sin \sqrt{t}\} = \frac{\sqrt{\pi}}{2s^{3/2}} e^{-1/4s}$, show that $L\left\{\frac{\cos \sqrt{t}}{\sqrt{t}}\right\} = \frac{\sqrt{\pi}}{s^{1/2}} e^{-1/4s}$.

Unit: I — transforms of elementary functions | **Marks:** 4 (was 6-7 in the 2023 papers) | **Section:** **Q1b** in Mar 24; **Q1c** in Jun 23. | **Appears alongside:** the t^n -multiplication proof — **always** (Law 1) | **History:** 2 of 4 with this exact wording; the $\sin \sqrt{t}$ family is 4 of 4 | **Note:** differentiate the given result with respect to t , or use the series for $\sin \sqrt{t}$

T1-14 Triangular wave — periodic function transform

Find the Laplace transform of the triangular wave function of period $2a$, given by $f(t) = \begin{cases} t, & \text{for } 0 \leq t \leq a \\ 2a - t, & \text{for } a \leq t \leq 2a \end{cases}$

Unit: I — transform of a periodic function | **Marks:** 7 | **Section:** the **d** part of whichever question does *not* carry the Law-1 pair. Q1c in Apr 23, Q2c in Jun 23. | **Appears alongside:** the other members of its question — typically a division-by- t evaluation and a definition | **History:** 2 of 4 papers with this exact wave; the periodic-function *topic* is 4 of 4 | **Note:** answer: $(1/s^2) \tanh(as/2)$. Sep 23 used the 2π sawtooth, Mar 24 the half-sine rectifier

§6 TIER 2 — certain to repeat, but with DIFFERENT VALUES

What qualifies: the **topic** has appeared in every paper (or 3 of 4), but the examiner changes the matrix, the function or the constants each time. You cannot memorise an answer — you must own the **method**. Each card below states the method checkpoint that actually earns the marks. **Marks covered by Tier 1 + Tier 2 together: 100 of 100 on every paper analysed.**

T2-01 QR factorization

Find the QR factorization of a 3×3 or 4×3 matrix. Mar 24 used $\begin{bmatrix} 3 & -5 & 1 \\ 1 & 1 & 1 \\ -1 & 5 & -2 \\ 3 & -7 & 8 \end{bmatrix}$ — Jun 23 used the 4×3 staircase

$\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ and Sep 23 the 3×3 staircase $\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$.

Unit: IV — QR-factorization | **Marks:** 10 | **Section:** Q8a in Mar 24 and Jun 23; Q7a in Sep 23; Q8b in Apr 23. | **Appears alongside:** the least-squares question (T1-02) — together they are all 20 marks of Q8 (Law 8) | **History:** 4 of 4 papers — matrix changes every single time | **Note:** Gram-Schmidt the columns, normalise to get Q , then $R = Q^T A$. Always state R is upper triangular

T2-02 Composition of matrix transformations + image of a point

Determine the matrix that describes a reflection in the x - (or y -) axis, followed by a rotation through θ , followed by a dilation or contraction of factor k . Find the image of a given point.

Mar 24: reflect in x , rotate $\pi/2$, contract $1/3$, point $\begin{bmatrix} 4 \\ 1 \end{bmatrix}$ • Sep 23: reflect in x , rotate $\pi/2$, dilate 3, point $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ • Jun 23: reflect in x , rotate $\pi/2$, contract $1/2$, point $\begin{bmatrix} 4 \\ -1 \end{bmatrix}$ • Apr 23: reflect in y , rotate $\pi/6$, dilate $3/2$, point $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$

Unit: III — composition of matrix transformations | **Marks:** 6 | **Section:** Q6a in Mar 24 and Sep 23; Q6c in Jun 23; Q5a in Apr 23. | **Appears alongside:** the transition matrix (T1-07) and the kernel&range question — the three-LOCK block (Law 6) | **History:** 4 of 4 papers — only the axis, angle, factor and point change | **Note:** multiply **right to left:** $k R_\theta \text{Ref}$. Reflection in x is $\text{diag}(1, -1)$; in y is $\text{diag}(-1, 1)$

T2-03 Kernel and range + verify rank-nullity

Determine the kernel and range of the transformation defined by a matrix, and hence verify the rank-nullity theorem.

Mar 24: $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & 1 \\ 1 & 1 & 4 \end{bmatrix}$ • Sep 23: a 3×4 matrix • Jun 23: $\begin{bmatrix} 1 & 2 \\ 3 & 0 \end{bmatrix}$ • Apr 23: $T(x, y) = (3x, x-y, y)$ given as a formula, not a matrix

Unit: III — kernel and range | **Marks:** 7 | **Section:** Q6c in Mar 24; Q6b in Jun 23; Q5c in Sep 23; Q6c in Apr 23. | **Appears alongside:** the composition and transition-matrix questions in the newest papers (Law 6) | **History:** 4 of 4 papers | **Note:** row-reduce, read $\text{nul}(A)$ for the kernel and pivot columns for the range, then show $\text{rank} + \text{nullity} = \text{number of columns}$. Be ready for the *formula* version too

T2-04 Prove a transformation is linear + find images

Prove that a given T is linear and find the images of two specified vectors.

Mar 24: $T: P_2 \rightarrow P_2$, $T(ax^2+bx+c) = (a+b)x^2+c$; image of $5x^2+6x+1$ • Apr 23 & Sep 23: $T(x, y) = (3x+y, 2y, x-y)$; images of $(1, 2)$ and $(2, -5)$ • Jun 23: $T(x, y) = (2x, x+y)$; images of $(1, 2)$ and $(-1, -4)$

Unit: III — linear transformations | **Marks:** 7 | **Section:** Q5b — in all four papers, without exception. | **Appears alongside:** the vector-space basis / span question at Q5a and the rotation proof at Q5c (Law 5) | **History:** 4 of 4 papers, always in the same slot | **Note:** check additivity and homogeneity separately, then substitute. The polynomial-space version is the newest twist — be ready for it

T2-05 Solve ODEs / simultaneous ODEs by Laplace transforms

Either a single linear ODE with initial conditions, or a pair of simultaneous ODEs.

Mar 24 Q4d: $dx/dt - 2y = \cos 2t$; $dy/dt + 2x = \sin 2t$, $x(0)=1$, $y(0)=0$ • Apr 23 & Jun 23: the “particle moving on a plane curve” version leading to $4x^2+4xy+5y^2 = 4$ • Jun 23 & Sep 23: $y'' + 4y' + 3y = e^{-t}$ • Apr 23: $y''' + 2y'' - y' - 2y = 0$

Unit: II — solution of linear and simultaneous ODEs | **Marks:** 7 | **Section:** the **d** part of Q3 or Q4 in the new format — and it sits with the Heaviside question (Law 3). | **Appears alongside:** the Heaviside piecewise question (T1-08) — **the 14-mark pairing** | **History:** 4 of 4 papers | **Note:** the $y''+4y'+3y = e^{-t}$ problem has been set twice with different initial conditions — do both versions

T2-06 Laplace transform of a periodic function

Find $L\{f(t)\}$ for a periodic f of period T , using $\frac{1}{1 - e^{-sT}} \int_0^T e^{-st} f(t) dt$.

Mar 24 Q2d: half-wave rectifier, $f(t) = E \sin(\omega t)$ on $0 < t < \pi/\omega$, 0 on $\pi/\omega < t < 2\pi/\omega$ • Apr 23 & Jun 23: triangular wave of period $2a$ (see T1-14) • Sep 23: $f(t) = t$ on $(0, \pi)$, $\pi - t$ on $(\pi, 2\pi)$

Unit: I — transform of a periodic function | **Marks:** 7 (+ a 2-mark “write the formula” version) | **Section:** the question *opposite* the Law-1 pair. Q2d in Mar 24, Q1c in Apr 23, Q2c in Jun 23, Q1b in Sep 23. | **Appears alongside:** a division-by- t or multiplication-by- t evaluation, plus the 2-mark definition | **History:** 4 of 4 papers | **Note:** Mar 24 also asked the **formula itself** as a 2-mark question (Q1a). Memorise the statement, not just the technique

T2-07 Complete solution of $Ax = b$

Find the complete solution — particular solution plus null-space basis.

Mar 24 Q7a: $x+3y+3z=1$, $2x+6y+9z=5$, $-x-3y+3z=5$ • Apr 23 Q8a: $A = \begin{bmatrix} 1 & 0 & 2 & 3 \\ 1 & 3 & 2 & 0 \\ 2 & 0 & 4 & 9 \end{bmatrix}$, $B = \begin{bmatrix} 2 \\ 5 \\ 10 \end{bmatrix}$ • Jun 23 Q7b: a 3×4 system in b_1, b_2, b_3 — “by choosing appropriate B ”

Unit: IV — complete solution to $Ax = b$ | **Marks:** 6 (was 10 in the 2023 papers) | **Section:** **Q7a** in Mar 24; Q7b in Jun 23; Q8a in Apr 23. | **Appears alongside:** the four-subspaces question (T1-03) and the orthogonal projection — all of Mar 24 Q7 | **History:** 3 of 4 papers | **Note:** always write the answer as $x = x_p + c_1 n_1 + \dots$ and state the consistency condition

T2-08 Short inverse Laplace transforms

A 4-mark (formerly 6-mark) inverse transform by partial fractions, shifting, or a standard form.

Mar 24 Q4b: $L^{-1}\left\{\frac{3s+2}{s^2-s-2}\right\}$ • Jun 23: $L^{-1}\left\{\frac{e^{-s}(s+2)}{(s+1)^2}\right\}$ and $\{\text{inv}\}\{\{\tan^{-1}(2/s)\}\}$, and $L^{-1}\left\{\frac{5s+3}{(s-1)(s^2+2s+5)}\right\}$ • Apr 23: $L^{-1}\left\{\frac{s^3+6s^2+12s+8}{s^6}\right\}$

Unit: II — inverse transforms | **Marks:** 4 | **Section:** the **b** part of Q3 or Q4 — both questions carry one. | **Appears alongside:** whatever else is in that question — you get one of these no matter which you pick | **History:** 4 of 4 papers | **Note:** guaranteed marks in **both** options, so it never affects your choice

T2-09 Evaluate an improper integral $\int_0^\infty$ using Laplace transforms

Mar 24 Q1d(i): $\int_0^\infty \frac{e^{-t} \sin t}{t} dt$ • Apr 23 Q2c(i): $\int_0^\infty \frac{e^{-3t} - e^{-2t}}{t} dt$ • Sep 23 Q1c: $\int_0^\infty t e^{-2t} \cos 3t dt$

Unit: I — evaluation of integrals by Laplace transforms | **Marks:** 7 (as one half of the **d** part) | **Section:** **Q1d** in Mar 24; Q2c in Apr 23; Q1c in Sep 23. | **Appears alongside:** a multiplication-by- t transform such as $L\{t \sin 3t \cos 2t\}$ — the two halves of one 7-mark part | **History:** 3 of 4 papers | **Note:** recognise the integral as $L\{f(t)\}$ evaluated at the s in the exponent — here $s=1$ gives $\pi/4$. With no exponential, put $s = 0$

T2-10 $L\{t e^{at} \text{ trig}\}$ and division by t

Mar 24 Q2c: (i) $L\{t e^{-4t} \sin 3t\}$ (ii) $L\left\{\frac{t - \sinh at}{t}\right\}$ • Sep 23 Q1a: $L\{t e^{-t} \sin 3t\}$, $L\{(1-e^t)/t\}$ • Jun 23

Q1a(ii): $L\{(\cos 4t - \cos 2t)/t\}$ • Apr 23 Q1b: $L\{2^t + (\cos 2t - \cos 3t)/t + t \sin t\}$

Unit: I — multiplication by t^n , division by t | **Marks:** 7 | **Section:** the **c** or **d** part of either Q1 or Q2 — it appears in some form in *both*. | **Appears alongside:** everything — this is the connective tissue of Unit I; you cannot avoid it | **History:** 4 of 4 papers | **Note:** the two workhorses: $L\{t f\} = -F(s)$ and $L\{f/t\} = \int_s^\infty F(u) du$

T2-11 Diagonalize a matrix and hence find A^n

Apr 23 Q9a: $A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$, find A^4 • Sep 23 Q9a: $A = \begin{bmatrix} 1 & -1 \\ 2 & 4 \end{bmatrix}$, find A^6 • Jun 23 Q10a(i): $A = \begin{bmatrix} -1 & 4 \\ 0 & 3 \end{bmatrix}$, find A^5

Unit: V — similarity and diagonalization; linear recurrence relations | **Marks:** 10 | **Section:** the 10-mark **c** part of whichever question does *not* carry orthogonal diagonalization. | **Appears alongside:** the SVD question (Law 12) — they share the same question in Jun 23 and Sep 23 | **History:** 3 of 4 papers — absent only from Mar 24 | **Note:** $A^n = PD^nP^{-1}$. Its absence from the newest paper makes it **more** likely to return, not less

T2-12 Linear combination / span / linear independence / basis

Mar 24 Q5a: is $\left\{ \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 3 & 4 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 2 \\ 1 & 2 \end{bmatrix} \right\}$ a basis of M_{22} ? • Sep 23: do $\{(1,2,3), (-1,-1,0), (2,5,4)\}$ span R^3 ? • Jun 23: is $(1,-2)$ a linear combination of $(2,4)$ and $(3,6)$? and check independence of $\{(1,0,1), (2,1,3), (-1,3,2)\}$ • Apr 23: is $(4,-4,6)$ a linear combination of $(1,2,-3)$, $(2,-4,6)$, $(-1,2,-3)$?

Unit: III — linear combination, span, basis and dimension | **Marks:** 6 | **Section:** Q5a in the three newest papers; Q6a in Apr 23. | **Appears alongside:** the “prove T is linear” question at Q5b (Law 5) | **History:** 4 of 4 papers | **Note:** the M_{22} matrix-space version is the newest twist — flatten each matrix to a 4-vector and test the 4×4 determinant

T2-13 Unitary matrix check

Mar 24 Q10b: is $A = \begin{bmatrix} 0 & i \\ -i & 0 \end{bmatrix}$ unitary? • Jun 23 Q10a(ii): is $A = \frac{1}{2} \begin{bmatrix} 1+i & 1-i \\ 1-i & 1+i \end{bmatrix}$ unitary?

Unit: V — unitary matrices | **Marks:** 5 | **Section:** the property-check block — Q10b in Mar 24. | **Appears alongside:** the positive-definite check (T1-05) — they are the 5+5 pair (Law 10) | **History:** 2 of 4 papers, but 1 of 1 under the current format | **Note:** show $A^*A = I$, where A^* is the conjugate transpose. Two minutes of work for 5 marks

T2-14 Quadratic form — remove cross-product terms / test definiteness

Mar 24 Q9a: express $Q(x) = x_1^2 - 8x_1x_2 - 5x_2^2$ as a quadratic form with no cross-product terms • Sep 23 Q10b(ii): is $3x^2 + 5y^2 + 3z^2 - 2yz + 2zx - 2xy$ positive definite?

Unit: V — quadratic forms | **Marks:** 5 | **Section:** Q9a in Mar 24; part of Q10b in Sep 23. | **Appears alongside:** the Markov steady-state question and the SVD (Law 12) — all of Mar 24 Q9 | **History:** 2 of 4 papers, and it is in the current format | **Note:** build the symmetric matrix, orthogonally diagonalise it, substitute $x = Py$. This is orthogonal diagonalization wearing a hat

T2-15 The 2-mark definitions

Write the Laplace transform of a periodic function • Define the Laplace transform of a function • Define the Dirac-delta function and sketch its graph • Define the unit step function and represent it graphically • Define a periodic function with an example.

Unit: I and II | **Marks:** $2 \times 2 = 4$ marks per paper | **Section:** the **a** part of Q1, Q2, Q3 and Q4 — every one of the four questions opens with one. | **Appears alongside:** everything — you get one free regardless of which question you choose | **History:** new in the March 2024 format; Jun 23 had one embedded in a 6-marker | **Note:** four marks for four sentences and two sketches. Also learn **existence conditions** and the **convolution statement** — both are in your syllabus and both are natural 2-markers

§7 TIER 3 — probable, and the ones that can ambush you

What qualifies: topics that appeared once, or that sit in your syllabus but have never been asked. Tier 1 and Tier 2 will carry you to a first class on their own. **This tier is what separates a 75 from a 95** — and it is where a paper can surprise you, because every one of these is a legitimate question the examiner is entitled to set.

T3-01 PCA — Principal Component Analysis

The biggest single risk on this sheet. PCA is named explicitly in Unit V of your syllabus and in CO5 (“obtain eigenvalue decomposition of a matrix and use it to study the concepts of SVD and PCA”), yet it does not appear in *any* of the four papers. Expect: form the covariance matrix of a small data set, find its eigenvalues and eigenvectors, identify the first principal component, and state the proportion of variance it explains.

Unit: V — principal component analysis | **Marks:** 5 (or half a 10-marker with SVD) | **Section:** the **a** or **b** part of Q9 or Q10 — wherever the two 5-mark slots are. | **Appears alongside:** SVD, almost certainly — they are the same machinery and the syllabus lists them in one breath | **History:** **0 of 4 papers** — but 1 of 1 syllabuses and 1 of 1 course outcomes | **Note:** cheapest insurance on this sheet: centre the data, compute $C = (1/(n-1))X^T X$, diagonalise. One hour of work

T3-02 Conic sections from a quadratic form

Classify a conic $ax^2 + bxy + cy^2 = d$ by diagonalising its symmetric matrix, and identify it as an ellipse, hyperbola or parabola in the rotated axes.

Unit: V — quadratic forms and conic sections | **Marks:** 5 | **Section:** the same 5-mark slot as the quadratic-form question (Mar 24 Q9a). | **Appears alongside:** the Markov / SVD block (Law 12) | **History:** **0 of 4 papers**, but explicitly in the syllabus | **Note:** the Mar 24 quadratic-form question is one sentence away from being this question. Learn the sign rule: both eigenvalues same sign → ellipse, opposite signs → hyperbola

T3-03 Gram-Schmidt — construct an orthonormal basis

Sep 23 Q8a: construct an orthonormal basis for $W = \text{span}\{x_1, x_2, x_3\} \subset \mathbb{R}^4$, where $x_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$, $x_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \end{bmatrix}$, $x_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}$, using the Gram-Schmidt process.

Unit: IV — orthonormal bases and Gram-Schmidt | **Marks:** 10 | **Section:** **Q8a** — the slot QR normally occupies. It is QR’s natural substitute. | **Appears alongside:** the least-squares question (it was Q8b in Sep 23, exactly as in Mar 24) | **History:** 1 of 4 papers — but you get it for free with QR | **Note:** if you can do QR you can already do this. Just stop before forming R

T3-04 Orthogonal projection of one vector onto another

Let $y = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$ and $u = \begin{bmatrix} 4 \\ -7 \end{bmatrix}$. Find the orthogonal projection of y onto u , and write y as the sum of two orthogonal vectors, one in $\text{span}\{u\}$ and one orthogonal to u .

Unit: IV — projections | **Marks:** 7 | **Section:** **Q7b** in Mar 24 — brand new in the current format. | **Appears alongside:** the complete solution at Q7a and the four-subspaces question at Q7c — all of Mar 24 Q7 | **History:** 1 of 4 papers, but 1 of 1 under the current syllabus, whose Unit IV is literally titled **Orthogonal Projections** | **Note:** $\text{proj} = ((y \cdot u)/(u \cdot u))u$. Five minutes for 7 marks — do not lose these

T3-05 Engineering application — LR circuit by Laplace transforms

A voltage Ee^{-at} is applied at $t = 0$ to a circuit of inductance L and resistance R . Show that the current at time t is $\frac{E}{R - aL}(e^{-at} - e^{-Rt/L})$, using Laplace transforms.

Unit: II — engineering applications | **Marks:** 7 | **Section:** **Q3d** in Mar 24. | **Appears alongside:** the Heaviside question at Q3c — this *is* the ODE half of the Law-3 pairing in that paper | **History:** 1 of 4 papers, but 1 of 1 under the current syllabus, which names “engineering applications” outright | **Note:** it is just $L di/dt + Ri = Ee^{-at}$, $i(0)=0$. If Law 3 holds, this or an ODE sits next to your Heaviside marks — learn it

T3-06 Markov matrix — steady-state vector

Find the steady-state solution of the Markov matrix $A = \begin{bmatrix} 0.95 & 0.03 \\ 0.05 & 0.97 \end{bmatrix}$

Unit: V — eigenvalue applications (not named in the syllabus text) | **Marks:** 5 | **Section:** Q9b in Mar 24. | **Appears alongside:** the quadratic form at Q9a and the SVD at Q9c | **History:** 1 of 4 papers — and it is the only borderline-syllabus item on this sheet | **Note:** solve $(A - I)x = 0$ and scale so the entries sum to 1. Fifteen minutes to learn, 5 marks

T3-07 Proofs you have seen once — and the two that are overdue

Seen once: $L\{f(t)/t\} = \int_0^\infty F(s) ds$ (Jun 23 Q2b, with $L\{\sin 4t/t\}$) • $L\{(\sinh at)f(t)\} = \frac{1}{2}[F(s-a) - F(s+a)]$ (Sep 23 Q2b, with $L[t^8 \sinh 2t]$) • state and prove the convolution theorem (Sep 23 Q3a).

Overdue — in the syllabus, never asked: existence conditions for the Laplace transform (a natural 2-marker)
• transform of **derivatives** stated as a proof • a direct **linear recurrence relation** (Fibonacci-style) solved by diagonalization.

Unit: I, II and V | **Marks:** 2 to 7 | **Section:** the **a** (2 m), **c** (7 m) or 10-mark slots depending on which appears. | **Appears alongside:** varies — proofs cluster with the other bookwork in whichever question carries them | **History:** 1 of 4 each, or 0 of 4 for the overdue ones | **Note:** the examiner has a stock of about six Unit-I proofs and rotates them. The t^n proof (T1-04) has come up four times; these are the alternates

§8 The exam-hall decision card

Your first three minutes: do not start writing. Read all ten questions and mark your five.

Each rule below tells you what to **look for**, not what to assume. The laws in §4 predict where things land, but the paper in front of you is the authority. Use the trigger.

Unit	Trigger — scan for this	Then answer	Why — and what you are giving up
I	Find the question containing the t^n-multiplication proof .	That question. It also holds the $\sin \sqrt{t} / \cos \sqrt{t}$ problem (Law 1, 4/4).	$7 + 4 = 11$ marks of pure recall in one question. You give up the periodic-function problem, which lands in the other one (Law 2) — so prepare it anyway, because if the proof is missing, the periodic question <i>is</i> your answer.
II	Find the question containing the Heaviside / unit-step piecewise function.	That question. An ODE, a system of ODEs, or the circuit application will be sitting next to it (Law 3, 4/4).	$7 + 7 = 14$ guaranteed marks , plus a 2-mark definition and a 4-mark inverse transform to complete the 20. You give up the convolution question, which sits opposite (Law 4).
III	Find the question containing the composition of transformations (reflection $\rightarrow$ rotation $\rightarrow$ dilation/contraction).	That question. In the two newest papers it also carried the transition matrix <i>and</i> the kernel&range question (Law 6).	$6 + 7 + 7 = 20$ marks from three LOCK topics . If the composition question is instead paired with something unfamiliar, fall back on the Q5 proof block: basis/span + prove- T -is-linear + rotation proof (Law 5).
IV	Compare the two shapes. Q7 is $6+7+7$; Q8 is $10+10$.	Q8 if it is QR + least-squares (Law 8, and what Mar 24 did). Q7 if you want the four-subspaces marks — that question is <i>always</i> Q7 (Law 7, 4/4).	$Q8 =$ two LOCK topics, but two long computations with no partial safety net. $Q7 =$ three shorter problems, more forgiving if one goes wrong. If your algebra is shaky under time pressure, take Q7.
V	Find the question containing orthogonal diagonalization .	That question. The property-check block (positive definite / Hermitian / unitary) travels with it (Laws 10 & 11).	$10 + 5 + 5 = 20$, where the two 5-markers are ten minutes of work between them. You give up SVD, which sits opposite (Law 9) — but SVD is a LOCK too, so prepare both.

Three ways this plan can fail — and the counter

- A law breaks.** Every law is 3/4 or 4/4, not 5/5. If the composition question is *not* next to the transition matrix, don't force it — count the marks you can actually secure in each question and take the larger number. The laws are a shortcut, not a substitute for reading the paper.
- A brand-new question appears.** March 2024 introduced three at once — the LR circuit, the orthogonal projection and the Markov steady state. That is 19 marks of novelty in a single paper. Tier 3 exists precisely for this; it is the reason the realistic ceiling is 95, not 100.
- PCA is set.** It is in your syllabus and your course outcomes and it has never been asked. If it appears it will be worth 5 marks in Unit V. One hour of preparation removes this risk entirely.

§9 Back-test — what this sheet would actually have scored

Every prediction sheet should be tested against the evidence it was built from. Below: for each paper, the questions the §8 decision card would have selected, and the marks that Tier 1 + Tier 2 preparation alone would have secured.

Paper	Questions selected	Tier 1+2	Score	Anything not covered
April 2023 Regular SEE	Q1, Q4, Q5, Q7, Q10	100 / 100	100	Nothing. Every part maps to a Tier 1 or Tier 2 card.
June 2023 Make-up	Q1, Q3, Q5, Q8, Q9	100 / 100	100	Nothing.
Aug/Sep 2023 Backlog	Q1, Q3, Q5, Q7, Q10	100 / 100	100	Nothing on the selected path. Had you taken Q8 instead of Q7 you would have met Gram-Schmidt (T3-03, 10 m).

March 2024 SEE — current format	Q1, Q4, Q6, Q8, Q10	100 / 100	100	Nothing on the selected path. Q3 would have cost you 7 marks (the LR circuit, T3-05) and Q7 would have cost 7 (the projection, T3-04).
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What that number does and does not mean

A 100/100 back-test is **partly circular** — the tiers were built from these four papers, so of course they cover them. What it honestly demonstrates is narrower but still useful: **the choice rules in §8 select a fully-covered question in every unit of every paper**, and the syllabus is small enough that roughly 30 distinct problem types exhaust it.

The forward-looking estimate is different, and here it is straight:

If you prepare...	Expected	Floor	Reasoning
Tier 1 only	55-65	45	63 marks of the March 2024 paper sat in Tier 1, but you cannot always reach all of it through the choice structure.
Tier 1 + Tier 2	80-92	70	Covers every topic that has appeared more than once. The floor assumes two brand-new questions land in your chosen sections, as happened in March 2024.
Tier 1 + 2 + 3	90-98	82	The only remaining exposure is a genuinely unprecedented question. Tier 3 is roughly six extra hours of work.

§10 Study order — highest marks per hour first

#	Do this	Marks defended	Rough hours	Why it is in this position
1	T1-04 t^n proof + T1-13 $\sin \sqrt{t}/\cos \sqrt{t}$	11	1.5	4/4 frequency, they sit in the same question (Law 1), and both are reproducible from memory. Nothing else on the paper is this reliable.
2	T1-08 Heaviside + T2-05 ODEs by Laplace transforms	14	3	The Law-3 pairing — 4/4, and the largest guaranteed block on the paper. Do the $t^2/4t/8$ function until it is automatic.
3	T1-01 orthogonal diagonalization + T1-05 positive definite + T2-13 unitary + T1-12 Hermitian	20	3	Laws 10 & 11 put all four in one question. The three property-checks are ten minutes of work for 10 marks — the best rate on the entire paper.
4	T2-02 composition + T1-07 transition matrix + T2-03 kernel & range	20	3.5	Three LOCK topics that formed one whole question in the two newest papers (Law 6). Memorise the transition-matrix problem exactly — same bases twice running.
5	T2-01 QR + T1-02 least-squares	20	4	Both 4/4 and paired in Q8 (Law 8). Slower than the blocks above because the arithmetic is long — but the least-squares data has repeated verbatim.
6	T1-03 four subspaces + T2-07 complete solution + T3-04 projection	20	3	Your Unit IV alternative (Q7). Cheap because the four-subspaces matrix has never changed. Gives you a genuine choice on the day.
7	T1-06 SVD + T2-11 diagonalize and find A^n	10	2.5	SVD is 4/4 but lands in the question you are <i>not</i> answering (Law 9). Insurance, not primary. A^n is overdue — absent from March 2024 after three straight appearances.
8	T2-06 periodic functions + T1-14 triangular wave + T2-15 the 2-mark definitions	11	2	Covers the Unit I question you did <i>not</i> choose, so nothing in that unit can hurt you. The definitions are four marks for four sentences.
9	T2-04 prove T linear + T2-12 span/basis + T1-10 rotation proof	20	2	The Q5 fallback block (Law 5). Fast, because it is mostly bookwork. Buy yourself a second option in Unit III.
10	T2-08 short inverse transforms + T2-09 improper integrals + T2-10 $t e^{at}$ / division by t + T1-09 , T1-11	22	3	The connective tissue of Units I and II. Individually small, collectively large, and they appear in <i>both</i> options of both units.
11	T3-01 PCA + T3-02 conic sections + T3-06 Markov	10–15	2	Do not skip this row. Three cheap topics that close your only real syllabus blind spots. PCA in particular is named in your course outcomes and has never been asked — which makes it overdue, not safe.
12	T3-03 Gram-Schmidt + T3-05 LR circuit + T3-07 the alternate proofs	10–20	2.5	Last, because each is one appearance in four papers. But Gram-Schmidt is free once you know QR, and the circuit problem sits right next to your 14-mark Heaviside block.

Rows 1–6 alone defend every unit and total roughly 18 hours. If you have three days, that is the plan. If you have a week, do all twelve.

§11 Appendix — complete question index of all four papers

Everything the prediction is built on, so you can check any claim yourself. **T** = the tier card that covers it.

Paper N — SEMESTER END EXAMINATIONS, MARCH 2024 (current format — plan against this one)

Course: Laplace Transforms and Vector Space · Code CS/IS/AI/AD/CY/CI31 · 100 marks / 3 hrs · This is the paper contained in *both* SEE_25.pdf and IS131_MATH_SEE_2024_1.pdf.

Q	Marks	Question	Tier card
1a	02	Write the Laplace transform of a periodic function.	T2-15
1b	04	Given $L\{\sin \sqrt{t}\} = \frac{\sqrt{\pi}}{2s^{3/2}} e^{-1/4s}$, show $L\{\cos \sqrt{t}/\sqrt{t}\} = \frac{\sqrt{\pi}}{s^{1/2}} e^{-1/4s}$.	T1-13
1c	07	Prove $L\{t^n f(t)\} = (-1)^n d^n\{F(s)\}/ds^n$.	T1-04
1d	07	Evaluate (i) $\int_0^\infty \frac{e^{-t} \sin t}{t} dt$ (ii) $L\{t \sin 3t \cos 2t\}$.	T2-09 / T2-10
2a	02	Define the Laplace transform of a function.	T2-15
2b	04	Obtain the Laplace transform of $(\sqrt{t} + 1/\sqrt{t})^3$.	T2-10
2c	07	Evaluate (i) $L\{t e^{-4t} \sin 3t\}$ (ii) $L\{\frac{t - \sinh at}{t}\}$.	T2-10
2d	07	Periodic function of period $2\pi/\omega$: $f(t) = F \sin(\omega t)$ on $0 < t < \pi/\omega$, 0 on $\pi/\omega < t < 2\pi/\omega$. Show $L\{f\} = E\omega/((s^2 + \omega^2)(1 - e^{-\pi s/\omega}))$.	T2-06
3a	02	Define the Dirac-Delta function and sketch its graph.	T2-15
3b	04	Evaluate $L^{-1}\{\log \frac{s^2+1}{s(s+1)}\}$.	T1-09
3c	07	Express $f(t) = t^2$ ($0 < t < 2$), $4t$ ($2 < t < 4$), 8 ($t > 4$) via the Heaviside function; find its LT.	T1-08
3d	07	LR circuit: voltage Ee^{-at} at $t=0$; show the current is $(E/(R-aL))(e^{-at} - e^{-Rt/L})$.	T3-05
4a	02	Define the unit step function and represent it graphically.	T2-15
4b	04	Find $L^{-1}\{\frac{3s+2}{s^2-s-2}\}$.	T2-08
4c	07	Verify the convolution theorem for $f_1(t) = t$ and $f_2(t) = \cos t$.	T1-11
4d	07	Solve $dx/dt - 2y = \cos 2t$; $dy/dt + 2x = \sin 2t$, $x(0)=1$, $y(0)=0$.	T2-05
5a	06	Is $\{\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 3 & 4 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 2 \\ 1 & 2 \end{bmatrix}\}$ a basis of M_{22} ?	T2-12
5b	07	$T: P_2 \rightarrow P_2$, $T(ax^2+bx+c) = (a+b)x^2+c$. Show linear; image of $5x^2+6x+1$.	T2-04
5c	07	Show $T(x) = Ax$ rotates x through θ if $A = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$.	T1-10
6a	06	Reflection in x -axis, then rotation $\pi/2$, then contraction $1/3$. Image of $\begin{bmatrix} 4 \\ 1 \end{bmatrix}$.	T2-02
6b	07	$B = \{(1,2), (3,-1)\}$, $B' = \{(3,1), (5,2)\}$. Transition matrix; $[u]_B = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, find $[u]_{B'}$.	T1-07
6c	07	Kernel and range of $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & 1 \\ 1 & 1 & 4 \end{bmatrix}$; verify rank-nullity.	T2-03
7a	06	Complete solution of $x+3y+3z=1$, $2x+6y+9z=5$, $-x-3y+3z=5$.	T2-07
7b	07	$y = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$, $u = \begin{bmatrix} 4 \\ -7 \end{bmatrix}$: orthogonal projection of y onto u ; split y into two orthogonal parts.	T3-04
7c	07	Dimension and basis for the four fundamental subspaces of $A = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{bmatrix}$.	T1-03

8a	10	QR factorization of $A = \begin{bmatrix} 3 & -5 & 1 \\ 1 & 1 & 1 \\ -1 & 5 & -2 \\ 3 & -7 & 8 \end{bmatrix}$.	T2-01
8b	10	Least-square solution of $AX=b$, $A = \begin{bmatrix} 1 & 3 & 5 \\ 1 & 1 & 0 \\ 1 & 1 & 2 \\ 1 & 3 & 3 \end{bmatrix}$, $b = \begin{bmatrix} 3 \\ 5 \\ 7 \\ -3 \end{bmatrix}$.	T1-02
9a	05	Express $Q(x) = x_1^2 - 8x_1x_2 - 5x_2^2$ with no cross-product terms.	T2-14
9b	05	Steady-state solution of the Markov matrix $A = \begin{bmatrix} 0.95 & 0.03 \\ 0.05 & 0.97 \end{bmatrix}$.	T3-06
9c	10	SVD of $A = \begin{bmatrix} 1 & 1 \\ 3 & -3 \end{bmatrix}$.	T1-06
10a	05	Is $A = \begin{bmatrix} 1 & -2 & 1 \\ -2 & 4 & -2 \\ 1 & -2 & 1 \end{bmatrix}$ positive definite?	T1-05
10b	05	Is $A = \begin{bmatrix} 0 & i \\ -i & 0 \end{bmatrix}$ unitary?	T2-13
10c	10	Orthogonally diagonalize $A = \begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix}$.	T1-01

Paper A — SEMESTER END EXAMINATIONS, APRIL 2023

Course: Linear Algebra and Laplace Transforms · Code CS/IS/AI/AD31 · older title, same five units

Q	Marks	Question	Tier card
1a	06	(i) $L\{\sqrt{t} - 1/\sqrt{t}\}^3$ (ii) $L\{\int_0^t e^{-t} \cos t \, dt\}$	T2-10
1b	07	$L\{2^t + (\cos 2t - \cos 3t)/t + t \sin t\}$	T2-10
1c	07	Triangular wave of period $2a$: t on $[0, a]$, $2a - t$ on $[a, 2a]$.	T1-14
2a	06	Find $L\{\sin \sqrt{t}\}$.	T1-13
2b	07	Prove $L\{t^n f(t)\} = (-1)^n d^n \{F(s)\} / ds^n$.	T1-04
2c	07	(i) $\int_0^\infty (e^{-3t} - e^{-2t})/t \, dt$ (ii) $L\{e^{-3t} t^8\}$	T2-09
3a	06	(i) $L^{-1}\{(s^3 + 6s^2 + 12s + 8)/s^6\}$ (ii) $L^{-1}\{(se^{-s/2} + \pi e^{-s})/(s^2 + \pi^2)\}$	T2-08
3b	07	$L^{-1}\{s^2/((s^2 + a^2)(s^2 + b^2))\}$ by convolution.	T1-11
3c	07	Particle on a plane curve: $y + 2x = \sin 2t$, $x - 2y = \cos 2t$; show $4x^2 + 4xy + 5y^2 = 4$.	T2-05
4a	06	$L^{-1}\{\log((s^2 + 1)/(s(s + 1)))\}$	T1-09
4b	07	Heaviside form of $t^2 / 4t / 8$ and its LT.	T1-08
4c	07	Solve $y'' + 2y' - y - 2y = 0$, $y(0) = 0$, $y'(0) = 0$, $y''(0) = 6$.	T2-05
5a	06	Reflection in y -axis, rotation $\pi/6$, dilation $3/2$. Image of $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$.	T2-02
5b	07	$T(x, y) = (3x + y, 2y, x - y)$: linear? Images of $(1, 2)$, $(2, -5)$.	T2-04
5c	07	Transition matrix from $B = \{(2, 3), (1, 2)\}$ to $B' = \{(1, 0), (0, 1)\}$; $u_B = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$.	T1-07
6a	06	Is $(4, -4, 6)$ a linear combination of $(1, 2, -3)$, $(2, -4, 6)$, $(-1, 2, -3)$?	T2-12
6b	07	Show $T(X) = AX$ rotates x through θ , $A = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$.	T1-10
6c	07	Range and kernel of $T(x, y) = (3x, x - y, y)$; verify rank-nullity.	T2-03
7a	10	Four fundamental subspaces of $A = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{bmatrix}$.	T1-03
7b	10	Least-squares for the 6×4 block-design matrix with $b = (-3, -1, 0, 2, 5, 1)$.	T1-02
8a	10	Complete solution: $A = \begin{bmatrix} 1 & 0 & 2 & 3 \\ 1 & 3 & 2 & 0 \\ 2 & 0 & 4 & 9 \end{bmatrix}$, $B = \begin{bmatrix} 2 \\ 5 \\ 10 \end{bmatrix}$.	T2-07

8b	10	QR factorization of $A = \begin{bmatrix} 1 & 3 & 5 \\ 1 & 1 & 0 \\ 1 & 1 & 2 \\ 1 & 3 & 3 \end{bmatrix}$.	T2-01
9a	10	Diagonalize $A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$; find A^4 .	T2-11
9b	10	(i) Is $\begin{bmatrix} 3 & 7-4i & -2+5i \\ 7+4i & -2 & 3+i \\ -2-5i & 3-i & 4 \end{bmatrix}$ Hermitian? (ii) Is $\begin{bmatrix} 1 & -2 & 1 \\ -2 & 4 & -2 \\ 1 & -2 & 1 \end{bmatrix}$ positive definite?	T1-12 / T1-05
10a	10	Orthogonally diagonalize $A = \begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix}$.	T1-01
10b	10	SVD of $A = \begin{bmatrix} 1 & -1 \\ -2 & 2 \\ 2 & -2 \end{bmatrix}$.	T1-06

Paper M — MAKE UP EXAMINATIONS, JUNE 2023

Course: Linear Algebra and Laplace Transforms · Code CS/IS/AI/AD31

Q	Marks	Question	Tier card
1a	06	(i) Define a periodic function with an example (ii) $L\{(\cos 4t - \cos 2t)/t\}$	T2-15 / T2-10
1b	07	Prove $L\{t^n f(t)\} = (-1)^n d^n\{F(s)\}/ds^n$.	T1-04
1c	07	Given $L\{\sin \sqrt{t}\}$, show $L\{\cos \sqrt{t}/\sqrt{t}\}$.	T1-13
2a	06	(i) $L\{e^{-2t}(2\cos 5t \sin 3t)\}$ (ii) $L\{\int_0^t te^{-t} \cos t \, dt\}$	T2-10
2b	07	Prove $L\{f(t)/t\} = \int_s^\infty F(s)ds$; hence $L\{\sin 4t/t\}$.	T3-07
2c	07	Triangular wave of period $2a$.	T1-14
3a	06	(i) $L^{-1}\{e^{-s}(s+2)/(s+1)^2\}$ (ii) $L^{-1}\{\tan^{-1}(2/s)\}$	T2-08
3b	07	$L^{-1}\{s^2/((s^2+a^2)(s^2+b^2))\}$ by convolution.	T1-11
3c	07	Solve $y''+4y'+3y = e^{-t}$, $y(0)=0$, $y'(0)=0$.	T2-05
4a	06	$L^{-1}\{(5s+3)/((s-1)(s^2+2s+5))\}$	T2-08
4b	07	Heaviside form of $\cos t / \cos 2t / \cos 3t$ on $(0, \pi)$, $(\pi, 2\pi)$, $(2\pi, \infty)$.	T1-08
4c	07	Particle on a plane curve $\rightarrow 4x^2+4xy+5y^2=4$.	T2-05
5a	06	Is $(1, -2)$ a linear combination of $(2, 4)$ and $(3, 6)$?	T2-12
5b	07	$T(x, y) = (2x, x+y)$: linear? Images of $(1, 2)$, $(-1, -4)$.	T2-04
5c	07	Transition matrix $B = \{(1, 2), (3, -1)\} \rightarrow B' = \{(1, 0), (0, 1)\}$; $U_B = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$.	T1-07
6a	06	Define linear dependence/independence; test $\{(1, 0, 1), (2, 1, 3), (-1, 3, 2)\}$.	T2-12
6b	07	Define kernel and range; verify rank-nullity for $\begin{bmatrix} 1 & 2 \\ 3 & 0 \end{bmatrix}$.	T2-03
6c	07	Reflection in x-axis, rotation $\pi/2$, contraction $1/2$. Image of $\begin{bmatrix} 4 \\ -1 \end{bmatrix}$.	T2-02
7a	10	Basis for $\text{col}(A)$ and $\text{nul}(A)$ of $A = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{bmatrix}$.	T1-03
7b	10	Complete solution of a 3×4 system, choosing an appropriate B .	T2-07
8a	10	QR factorization of $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$.	T2-01
8b	10	Least-squares: $A = \begin{bmatrix} 1 & 3 & 5 \\ 1 & 1 & 0 \\ 1 & 1 & 2 \\ 1 & 3 & 3 \end{bmatrix}$, $b = \begin{bmatrix} 3 \\ 5 \\ 7 \\ -3 \end{bmatrix}$.	T1-02

9a	10	Orthogonally diagonalize $A = \begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix}$.	T1-01
9b	10	(i) Is $\begin{bmatrix} 0 & 2+i & 1 \\ 2-i & i & 0 \\ 1 & 0 & 1 \end{bmatrix}$ Hermitian? (ii) Is $\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$ positive definite?	T1-12 / T1-05
10a	10	(i) Diagonalize $\begin{bmatrix} -1 & 4 \\ 0 & 3 \end{bmatrix}$, find A^5 (ii) Is $\frac{1}{2} \begin{bmatrix} 1+i & 1-i \\ 1-i & 1+i \end{bmatrix}$ unitary?	T2-11 / T2-13
10b	10	SVD of $A = \begin{bmatrix} 1 & 1 \\ 3 & -3 \end{bmatrix}$.	T1-06

Paper S — BACKLOG SUBJECT EXAMINATIONS, AUGUST / SEPTEMBER 2023

Course: Linear Algebra and Laplace Transforms · Code CS/IS/AI/AD31

Q	Marks	Question	Tier card
1a	06	(i) $L\{te^{-t}\sin 3t\}$ (ii) $L\{(1-e^t)/t\}$	T2-10
1b	07	Periodic function of period 2π : t on $(0, \pi)$, $\pi - t$ on $(\pi, 2\pi)$.	T2-06
1c	07	Evaluate $\int_0^\infty te^{-2t}\cos 3t \, dt$.	T2-09
2a	06	Show $L\{\sin \sqrt{t}\} = (\sqrt{\pi}/2s^{3/2})e^{-1/4s}$.	T1-13
2b	07	Prove $L\{(\sinh at)f(t)\} = \frac{1}{2}[F(s-a) - F(s+a)]$; evaluate $L[t^8 \sinh 2t]$.	T3-07
2c	07	Prove $L\{t^n f(t)\} = (-1)^n d^n[F(s)]/ds^n$.	T1-04
3a	06	State and prove the convolution theorem.	T1-11 / T3-07
3b	07	Heaviside form of $1/t^2$ on $(0,1]$, $(1,2]$, $(2,\infty)$.	T1-08
3c	07	Solve $dx/dt - y = e^{-t}$, $dy/dt + x = \sin t$, $x(0)=1$, $y(0)=0$.	T2-05
4a	06	$L^{-1}\{\log((s^2+9)/(s(s+9)(s-9)))\}$	T1-09
4b	07	Solve $y'' + 4y' + 3y = e^{-t}$, $y(0)=y'(0)=1$.	T2-05
4c	07	$L^{-1}\{s^2/((s^2+1)(s^2+4))\}$ by convolution.	T1-11
5a	06	Do $\{(1,2,3), (-1,-1,0), (2,5,4)\}$ span R^3 ? Express $(1,3,-2)$.	T2-12
5b	07	$T(x,y)=(3x+y, 2y, x-y)$: linear? Images of $(1,2)$, $(2,-5)$.	T2-04
5c	07	Kernel and range dimensions for $\begin{bmatrix} 1 & -2 & 3 & 5 \\ 1 & -1 & 8 & 7 \\ 2 & -4 & 6 & 10 \end{bmatrix}$.	T2-03
6a	06	Reflection in x-axis, rotation $\pi/2$, dilation 3. Image of $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$.	T2-02
6b	07	State and verify rank-nullity for $T(x,y)=(3x, x-y, y)$.	T2-03
6c	07	$B=\{(1,2),(3,-1)\} \rightarrow B'=\{(3,1),(5,2)\}$; $u_{B'}=\begin{bmatrix} 2 \\ 1 \end{bmatrix}$.	T1-07
7a	10	QR factorization of $A=\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$.	T2-01
7b	10	Basis and dimension of $\text{nul}(A)$ and $\text{col}(A)$ for $\begin{bmatrix} -3 & 9 & -2 & -7 \\ 2 & -6 & 4 & 8 \\ 3 & -9 & -2 & 2 \end{bmatrix}$.	T1-03
8a	10	Gram-Schmidt orthonormal basis for $\text{span}\{(1,1,1,1), (0,1,1,1), (0,0,1,1)\}$.	T3-03
8b	10	Least-squares: $A=\begin{bmatrix} 4 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix}$, $b=\begin{bmatrix} 2 \\ 0 \\ 11 \end{bmatrix}$.	T1-02
9a	10	Diagonalize $A=\begin{bmatrix} 1 & -1 \\ 2 & 4 \end{bmatrix}$; find A^6 .	T2-11
9b	10	SVD of $A=\begin{bmatrix} 4 & 11 & 14 \\ 8 & 7 & -2 \end{bmatrix}$.	T1-06

10a	10	Orthogonally diagonalize $A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$.	T1-01
10b	10	(i) Is $\begin{bmatrix} 3 & 7-4i & -2+5i \\ 7+4i & -2 & 3+i \\ -2-5i & 3-i & 4 \end{bmatrix}$ Hermitian? (ii) Is $3x^2+5y^2+3z^2-2yz+2zx-2xy$ positive definite?	T1-12 / T2-14

24CS31 — SEE Prediction Sheet. Built by transcribing four previous question papers question-by-question from the page images, cross-tabulating all 40 questions by topic and question slot, and filtering the result against the official 24CS31 syllabus. Frequencies, pairing laws and the back-test are computed from that table and are reproducible from §11. Predictions are inferences from four papers — a strong pattern, not a guarantee. Study the syllabus, use this to prioritise.